Clinical Medicine and Immunology: The Wisdom of Life That Institutionalizes Error

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Clinical Medicine and Immunology: The Wisdom of Life That Institutionalizes Error

1. Purpose and Questions of the Investigation

Figure 1: Clinical Medicine and Immunology: The Wisdom of Life That Institutionalizes Error

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This report presents the results of an investigation into whether theories in clinical medicine and immunology correspond structurally to the five-stage model of creation: Field ( ), Wave ( ), Edge ( ), Vortex ( ), and Bundle ( ).

The immune system is a system that "institutionally produces, selects, and remembers error." Molecular mechanisms such as V(D)J recombination and somatic hypermutation are mechanisms of "planned error production" that intentionally cut and rejoin DNA to generate diversity. Clinical medicine further encompasses a wide range of processes, including wound healing, fracture repair, cancer immunity, sepsis, autoimmunity, immunotherapy, and transplant tolerance, thereby providing a comprehensive view within a single domain of normal, pathological, and interventional variations of the five stages.

This document is an investigative report and does not aim to justify any particular theory. It reports, with clear distinctions, where structural correspondence was confirmed, where correspondence was weak, and what questions remain unresolved.

2. Methods of Investigation

Objects of Evaluation

Ten theories and clinical processes from the domains of clinical medicine and immunology were selected for evaluation. The selection criteria were as follows:

- A stepwise pattern of structural change is established in immune responses, tissue repair, or pathogenesis
- Molecular-level mechanisms are supported by the literature
- Clinical observations or experimental data are available

The specific objects of evaluation were adaptive immunity (clonal selection theory, the germinal center reaction), tissue repair (the active resolution of acute inflammation, wound healing, fracture repair), pathological processes (cancer immunoediting, sepsis-induced immunoparalysis, autoimmune disease), and immunological interventions (allergen immunotherapy and tolerance induction in transplantation immunity).

Evaluation Method

For each theory, correspondence with each stage of the five-stage model, namely Field, Wave, Edge, Vortex, and Bundle, was examined individually. With particular regard to Edge, emphasis was placed on its fit with three conditions: relational network (whether multiple elements are interacting), indeterminacy (whether the outcome remains unsettled), and vortex connectivity (whether the interaction at Edge drives the next stage).

For each theory, structural correspondence was examined from multiple independent perspectives, including immunology, molecular biology, clinical medicine, and pharmacology, and the strength of that correspondence was then assessed.

Criteria for Judgment

- **Strong structural correspondence**: Clear correspondence is confirmed for each of the five stages, and the driving mechanisms between stages can also be described at the molecular level
- **Partial correspondence**: Correspondence is confirmed for some of the five stages, but not across the entire sequence
- **Conditional correspondence**: Structural correspondence is recognizable, but leaves substantial room for interpretation, or the scope of the correspondence is constrained

Limits of the Investigation

This investigation examines whether structural similarity is present; it does not prove causal relations. Nor does it cover every theory in immunology, and there is a tendency for theories with more legible correspondences to the five-stage model to be selected. It should also be noted that applying the five stages to pathological processes admits a wider range of interpretation than their application to normal processes.

3. Overview of the Five-Stage Model

This section explains the five-stage model used as the reference framework in this investigation. It is a model that describes the structure of creative processes in stages and is not limited to any particular scale or domain.

Field: an undifferentiated state. It is an initial condition in which neither direction nor structure has yet been determined, and a potential space that serves as the precondition for change.

Wave: the stage of exploration in which multiple directions diverge and compete. Differences emerge within what was previously undifferentiated, and conflict and fluctuation appear.

Edge: a state of tension in which opposing elements coexist without converging on either side. This is the core stage of the model: through the interaction of different elements within a relationship, the transition to the next stage is prepared.

Vortex: the stage in which a new coherence (order) spontaneously arises out of tension. The oppositions sustained at Edge are reorganized into a higher-order structure through positive feedback. In clinical medicine, this is observed as clonal expansion, granulation tissue formation, callus formation, or the rise of immune-escape clones. A Vortex does not necessarily produce a constructive outcome; pathological proliferation and runaway inflammatory responses also operate in the structure of a Vortex.

Bundle: the stage in which form is fixed and stabilized as a reusable structure. In clinical medicine, this is realized as immune memory, tissue remodeling, long-term immune tolerance, or the establishment of disease. A Bundle is not an endpoint, but a precondition for the Field of the next cycle.

If any of these definitions feels discordant, that is itself an important perspective for this investigation. It is meaningful either to continue reading while holding that sense of discordance or to step away from the report at this point.

4. Findings: Overall View

Figure 2: Clinical Medicine and Immunology: The Wisdom of Life That Institutionalizes Error — ×5

Evaluation of the ten theories and clinical processes showed structural correspondence with the five-stage model in all ten cases. Eight displayed strong correspondence with Edge, which is the highest level observed across the investigation of thirty domains. The reason for this high rate of correspondence is that intercellular interactions in the immune system make the concrete reality of Edge comparatively easy to observe directly.

An overall view of the correspondences is presented below.

Theory / Process	Strength of correspondence	Correspondence of Edge	Special note
Clonal selection theory	Strong structural correspondence	Strong	Core principle of immunology; stage correspondence is natural
Cancer immunoediting (3E)	Strong structural correspondence	Strong	The Equilibrium phase is a long-duration state of Edge
Active resolution of acute inflammation	Strong structural correspondence	Strong	Stabilization at Bundle is an active program
Wound healing	Strong structural correspondence	Strong	A textbook example of the five stages
Sepsis and immunoparalysis	Conditional correspondence	Partial	Describes a failure mode of the five stages
Fracture repair	Strong structural correspondence	Strong	The qualitative phase transition from cartilage to bone is a distinctive element
Allergen immunotherapy	Strong structural correspondence	Strong	A clinical example of rewriting Bundle
Transplantation immunity and tolerance induction	Strong structural correspondence	Strong	Plasticity of the self/non-self boundary
Autoimmune disease	Conditional correspondence	Partial	Application to pathological processes requires caution
Germinal center reaction	Strong structural correspondence	Strong	Microcycles of the five stages recur in circulation

The most distinctive feature of the clinical medicine domain is that diverse variations of the five stages coexist within a single domain. Normal five-stage processes (clonal selection, tissue repair), pathological five-stage processes (cancer immune escape, sepsis, autoimmunity), and interventional five-stage processes (immunotherapy, tolerance induction) exist in parallel. This diversity suggests that the five-stage model is not a norm guaranteeing "good creation," but a value-neutral framework for describing the structure of creative processes themselves.

5. Findings: Domain-Specific Insights

5-1. Clonal Selection Theory and the Germinal Center Reaction: Planned Error Production

Clonal selection theory is the core principle of the adaptive immune response. First proposed by Jerne (1955) and elaborated by Burnet (1957), the theory holds that lymphocyte clones matching an antigen are selectively expanded, thereby generating an effector response and immune memory. Tonegawa (1983) clarified the molecular mechanism of V(D)J recombination and showed that receptor diversity on the order of 10^11 is generated through the intentional cutting and rejoining of DNA.

The structure of "generation of diversity -> selection -> memory" corresponds to the five-stage model as follows. Field corresponds to the diverse antigen receptor repertoire, the potential space already generated by V(D)J recombination; Wave corresponds to the manifestation of differential reactivity among specific clones following antigen invasion; Edge corresponds to interactions among antigen-presenting cells such as dendritic cells or macrophages and T cells and B cells; Vortex corresponds to clonal expansion and differentiation, namely the rise of effector cells; and Bundle corresponds to immune memory, that is, the long-term retention of memory T and B cells.

What is especially noteworthy about Edge is that the physical contact between the MHC-peptide complex and its receptor can be directly observed as the concrete basis of Edge as "relational selection." Furthermore, even in response to the same antigen, differentiation may proceed toward distinct directions such as Th1, Th2, or Th17 depending on microenvironmental conditions. The fact that the outcome is not yet fixed, that is, that indeterminacy remains present, corresponds structurally to the definition of Edge.

The germinal center reaction (Victora & Nussenzweig, 2012) is a finer-grained expansion of clonal selection. In germinal centers, B cells repeatedly cycle through "mutation -> selection": in the dark zone they undergo somatic hypermutation (SHM), and in the light zone they are selected by T follicular helper cells (Tfh). This cyclic re-entry is the most concrete example at the cellular level of the repeated microcycle of the five-stage model. After each cycle, BCR affinity increases and the process does not return to its point of departure; in other words, it has a spiral structure.

The distinctive feature of the germinal center is the spatial separation of the dark zone, where mutation is introduced, and the light zone, where selection occurs. In many processes, the five stages proceed temporally, but in the germinal center B cells move physically in order to transition between stages. This finding, that a temporal stage structure can be spatially separated and yet coexist, suggests an expanded range of application for the five-stage model.

5-2. Wound Healing and Fracture Repair: Stepwise Tissue Reconstruction

Wound healing is a tissue response that proceeds through the sequence of hemostasis -> inflammation -> cellular proliferation -> remodeling (Guo & DiPietro, 2010). Because the temporal ordering of these stages and their driving mechanisms have been clarified at the molecular level, this is one of the processes in which correspondence with the five-stage model can be confirmed most clearly.

Field corresponds to the wound bed and the hemostatic clot, a reservoir of fibrin and growth factors such as PDGF and TGF-b; Wave corresponds to the infiltration of inflammatory cells; Edge corresponds to multicellular interactions centered on macrophage phenotypic conversion from the inflammatory M1 state to the reparative M2 state; Vortex corresponds to the positive-feedback amplification of granulation tissue formation, angiogenesis, and re-epithelialization; and Bundle corresponds to remodeling, including collagen realignment, cross-linking, and wound contraction.

Of particular note here is macrophage phenotypic conversion (Landen et al., 2016). The same immune cell undergoes a qualitative shift from a pro-inflammatory type (M1) to a repair-promoting type (M2) in response to changes in the microenvironment, and this shift drives the next stage, namely granulation tissue formation as Vortex. At the same time, it has been pointed out that the M1/M2 dichotomy is a model representing the two ends of a continuous spectrum, and that actual macrophage phenotypes can be more diverse and may take mixed states (Murray et al., 2014). Even with this caveat, the structure remains intact in which a qualitative conversion responsive to the microenvironment drives transition between stages.

Chronic wounds, including diabetic ulcers, are states in which this conversion fails. Qualitative conversion does not occur at Edge and the process cannot reach Vortex; this can be understood as a clinical example of stagnation and failure within the five stages.

Fracture repair (Einhorn & Gerstenfeld, 2015) shares with wound healing the isomorphic pattern of "injury -> inflammation -> repair -> stabilization," while adding the qualitative phase transition of "cartilage -> bone." In endochondral ossification, the cartilage template is calcified and broken down, then replaced by bone produced by osteoblasts. This process, in which the original structure is created by passing through the destruction of a temporary structure, displays a double structure of the five stages: a provisional Bundle is destroyed and becomes a new Field, from which the original Bundle is then formed.

Moreover, the finding that remodeling at Bundle is optimized over years through interaction with the mechanical environment in accordance with Wolff's law shows that Bundle is not simply finished once completed; rather, its quality is determined within its relation to the environment.

5-3. Cancer Immunoediting: The Multi-Year Persistence of Edge

The 3E model of cancer immunoediting (Dunn et al., 2004) describes the coevolution of the immune system and tumors in three stages: Elimination, Equilibrium, and Escape. The immune system recognizes the tumor and attempts to eliminate it, while the tumor undergoes selection under immune pressure, ultimately establishing immune-escape clones. The clinical success of immune checkpoint blockade therapy (Sharma & Allison, 2015) supports this model.

In correspondence with the five stages, Field is the coexisting state of genetic diversity within the tumor and immune surveillance mechanisms; Wave is the manifestation of difference through neoantigen recognition; Edge is the interaction between immune cells and tumor cells; Vortex is the selective proliferation of immune-escape clones; and Bundle is fixation as an advanced tumor, or alternatively elimination and remission through immunotherapy.

What is especially striking is the Equilibrium phase. During this period, the immune system cannot eliminate the tumor, and the tumor cannot completely evade the immune system either; a sustained indeterminate state, fixed neither as elimination nor as escape, continues over years. This prolonged unresolved state is one of the most dramatic examples of the definition of Edge: a state of tension in which opposing elements coexist without converging on either side.

The immune-escape clone that rises as Vortex is a harmful "creation" from the standpoint of the organism. This fact shows that the five-stage model does not describe "good creation," but

rather the structure of creative processes as such, a value-neutral structure.

5-4. Active Resolution of Acute Inflammation: The Beginning Programs the Ending

The resolution of acute inflammation was traditionally thought to "subside naturally." However, research by Serhan and colleagues from the 2000s onward showed that it is an active program mediated by lipid mediators such as resolvins, protectins, and maresins (Serhan, 2014). Resolution is not the consequence of neglect; it is a process actively driven by dedicated molecular mechanisms.

In correspondence with the five stages, Field is tissue homeostasis; Wave is activation of the inflammatory cascade by injury or infection; Edge is the cytokine interaction among immune cells, endothelium, and epithelium, including neutrophil apoptosis and macrophage efferocytosis; Vortex is the rise of a local defense program; and Bundle is return to homeostasis through active resolution.

What is especially important here is the insight that "the beginning programs the ending" (Serhan & Savill, 2005). At the onset of inflammation, the resolution program is activated simultaneously. During the Field -> Wave transition, the return route toward Bundle is already being programmed. Biologically, this is evidence that the five stages are not a simple linear progression.

In addition, the finding that failure of the resolution program is directly linked to chronic inflammation, fibrosis, atherosclerosis, and related pathologies shows that stabilization at Bundle is not "fixation by inertia," but an active program.

5-5. Plasticity of Boundaries: Immunotherapy and Transplant Tolerance

Allergen immunotherapy (Akdis & Akdis, 2014) is a treatment that suppresses allergic reactions by converting an IgE-dominant immune response into an IgG4-dominant one through repeated low-dose exposure and by inducing regulatory T cells (Treg). With a history of more than one hundred years (Noon, 1911), both sublingual and subcutaneous immunotherapy have demonstrated long-term remission (Durham et al., 1999).

This treatment can be understood as a process that overwrites an inappropriate immune memory formed during initial sensitization, namely a Bundle that had once been completed, through a new five-stage cycle. The sensitized immune system becomes a new Field; therapeutic allergen administration produces a controlled Wave; a shift in immune balance occurs at Edge centered on Treg induction; proliferation of IgG4-producing B cells unfolds as Vortex; and long-term immune tolerance is established as a new Bundle. This is the clearest clinical evidence that a Bundle, once completed, is not irreversible but can be rewritten.

In transplantation immunity, Medawar (1944/1960) clarified the basic mechanisms of immune rejection against grafts, and Billingham, Brent & Medawar (1953) experimentally demonstrated that acquired immune tolerance can be induced. More recently, Kawai et al. (2008) reported cases in which grafts were maintained without immunosuppressive agents, namely clinical tolerance induction.

The most suggestive feature of transplantation immunity is the structure in which diametrically opposed Vortices bifurcate from the same Edge. The same event, namely host T-cell recognition of non-self, can branch toward rejection, the Vortex of effector T-cell infiltration, or toward tolerance, the Vortex of Treg expansion. This is one of the most dramatic examples of the indeterminacy of Edge.

Autoimmune disease (Rosenblum et al., 2015) shows plasticity in the opposite direction from transplant tolerance. Through combinations of genetic predisposition and environmental triggers, including infection and the gut microbiota, the immune system, which should not attack the

self, acquires self-reactivity. Molecular mimicry, the structural similarity between pathogen antigens and self-antigens, is one of the principal pathways of activation. Autoimmunity can be positioned as a pathological process of the five stages in which a once-correct discrimination at Edge deteriorates, but pathways of onset differ by disease, and there are clear limits to generalization.

Allergen immunotherapy, which rewrites Bundle; transplant tolerance induction, which accepts non-self and thus expands the boundary; and autoimmunity, which misrecognizes the self and thus contracts the boundary: what these three show is that the boundary between self and non-self is not fixed, but dynamically constituted and reconstituted according to conditions.

6. Cross-Cutting Patterns

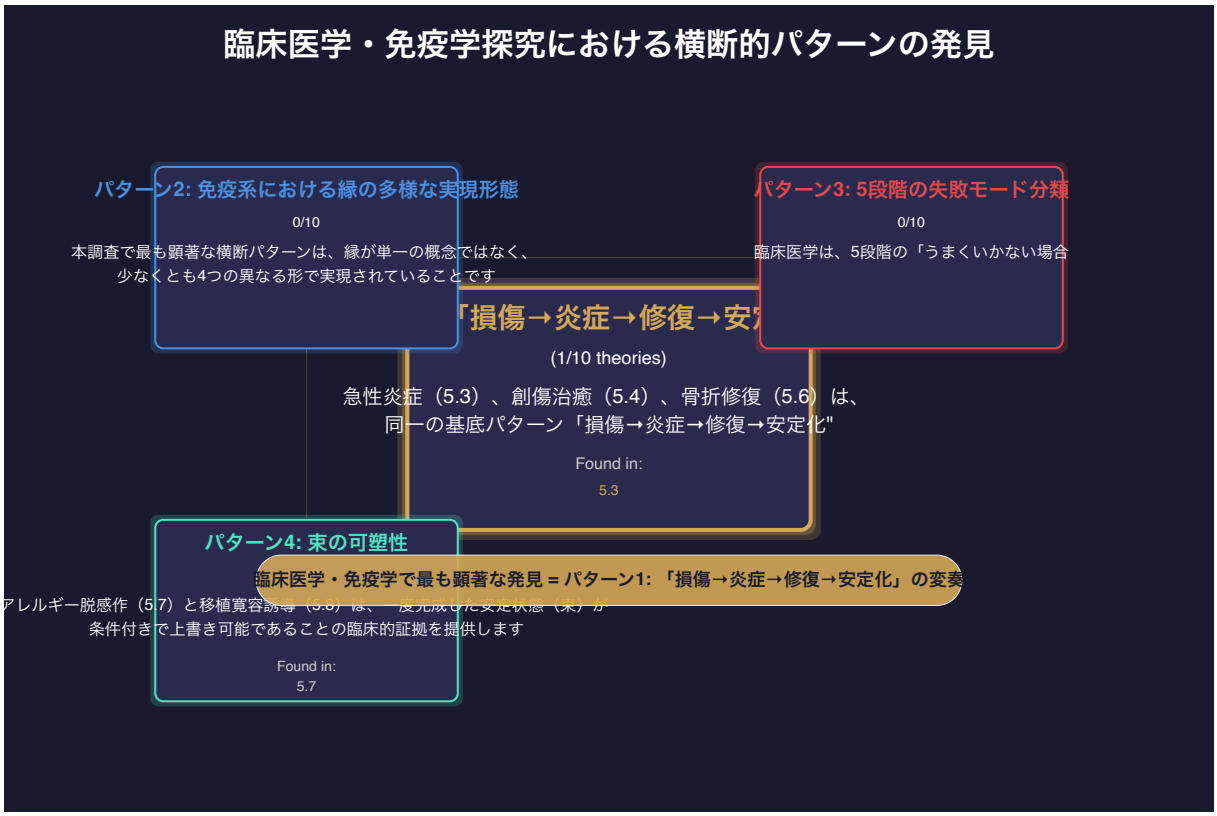

Figure 3: Clinical Medicine and Immunology: The Wisdom of Life That Institutionalizes Error

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This section reports structural patterns that recurred across the domain of clinical medicine and immunology beyond any individual theory.

The Abundance of Edge in the Immune System

Strong correspondence with Edge was confirmed in 8 of the 10 cases. This is the highest level observed among the thirty domains investigated. Intercellular interactions in the immune system, whether contact between antigen-presenting cells and T cells, selection between Tfh cells and B cells, macrophage phenotypic conversion, or confrontation between immune cells and tumor cells, are all describable as states in which different elements interact within a relation and the outcome remains unsettled. The immune system is a rare domain in which the concrete reality of Edge can be observed at the molecular level.

At the same time, the same label, Edge, is being used across different contexts, including Edge

as contact signaling, Edge as the self/non-self boundary, and Edge in therapeutic intervention. This suggests the need for a more refined sub-classification.

Coexistence of Normal, Pathological, and Interventional Variations

Clinical medicine is a domain in which the same five-stage structure appears as normal processes, such as clonal selection and tissue repair, pathological processes, such as cancer immune escape, sepsis, and autoimmunity, and processes shaped by deliberate intervention, such as immunotherapy and tolerance induction. The coexistence of these three kinds of variation within a single domain provides strong evidence that the five-stage model is a value-neutral framework for describing the structure of creative processes.

Repetition of the Pattern "Injury -> Inflammation -> Repair -> Stabilization"

The pattern "injury -> inflammation -> repair -> stabilization," common to wound healing, the resolution of acute inflammation, and fracture repair, recurs isomorphically from the molecular scale to the tissue scale. This repetition shows that the five-stage model is not limited to a specific scale, but can be reproduced as an isomorphic structure across different scales.

The Rewritability of Bundle

Allergen immunotherapy and transplant tolerance induction show clinically that a completed Bundle, whether immune memory or a classification of self and non-self, can be rewritten. Bundle is not an endpoint but a precondition for the Field of the next cycle. However, rewriting Bundle requires sustained intervention over years, immunotherapy typically taking 3 to 5 years, and whether all Bundles are rewritable in the same way remains an unresolved question.

7. Unresolved Questions

- **Limits of applying the five stages to pathological processes**: Sepsis-induced immunoparalysis is a "loss of responsiveness of the system as a whole," and thus includes the breakdown of the very condition presupposed by the five-stage model, namely that the system is responding. Where the limits of applicability lie is an important question for determining the scope of the theory.
- **Need for a sub-classification of Edge**: Edge as contact signaling, Edge as the self/non-self boundary, and Edge as therapeutic intervention are being managed under the same label. Developing a sub-classification that organizes their relations is a task for future work.
- **Conditions for the plasticity of Bundle**: Immunotherapy shows that rewriting Bundle is possible, but it remains unclear under what conditions Bundle can be rewritten and what the minimal conditions required for such rewriting are.
- **The theoretical positioning of the spatial five stages in the germinal center**: The light-zone/dark-zone structure of the germinal center shows that the five stages can be spatially separated and coexist. How this "spatialization of the five stages" should be situated theoretically remains unresolved.
- **Validity of the SIRS/CARS model**: The sequential model of sepsis as SIRS -> CARS is a simplification, and in reality both have been shown to progress simultaneously from early phases. The limits of this model may also affect its correspondence with the five stages.

8. Conclusion

Clinical medicine and immunology constitute a domain in which structural correspondence with the five-stage model was confirmed in all ten cases, with strong correspondence to Edge recognized

in eight. This represents one of the highest levels of structural correspondence among the thirty domains investigated.

The most robust finding is that the immune system incorporates the institutional production of error. Planned generation of diversity through V(D)J recombination and somatic hypermutation, relational selection through clonal selection, and long-term retention through immune memory: these three show a highly determinate structural correspondence with the five-stage model as Field, the pool of diversity -> Wave, the manifestation of difference -> Edge, relational selection -> Vortex, proliferation -> Bundle, memory.

Another powerful finding is that the five-stage model is a value-neutral structural description. Clinical medicine shows that normal, pathological, and interventional variations can all be described through the same structure, suggesting that this model is a framework for grasping the structure of creation rather than the quality of what is created.

As a more tentative finding, there is the dynamic constitution of boundaries and the rewritability of Bundle. Grounded in the clinical record of immunotherapy and tolerance induction, these insights support the cyclical structure of the five-stage model, Bundle -> new Field, though as interpretations of structural correspondence they still leave room for further verification.

An unresolved question remains the limit of applying the five stages to pathological processes. How far the five-stage model can describe states in which the responsiveness of the system itself is lost, as in sepsis-induced immunoparalysis, is an important question in determining the scope of the model.